O-T-370
November 21, 1962
(See 6.5)

FEDERAL SPECIFICATION

THIONYL CHLORIDE, REAGENT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one grade of thionyl chloride for use as a chemical reagent.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-C-300—Chemicals, Liquid; Packaging and Packing Of.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical requirements. The material shall conform to the chemical requirements shown in table I when tested as specified therein:

TABLE I. *Chemical requirements*

Requirements	Maximum	Minimum	Test method
	Percent	*Percent*	
Iron (calculated as $FeCl_2$)	0.10	—	4.6.1
Sulfur chloride (calculated as S_2Cl_2)	1.0	—	4.6.2
Sulfuryl chloride	5.0	—	4.6.3
Sulfur dioxide	2.0	—	4.6.4
Purity (see 3.2)	—	94.0	4.6.5

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3.2 **Purity.** The material shall be no less than 94.0 percent pure, based on total sulfur content corrected for sulfur chloride and sulfuryl chloride or by difference, whichever shall be the lower.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of section 5 have been satisfied.

4.3 **Lotting.** A lot shall consist of the thionyl chloride offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials provided the operation is continuous. In the event that the process is a batch operation, each batch shall continue as a lot. (See. 6.3.)

4.4 **Sampling.**

4.4.1 *For nondestructive examination.* Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2 *For test.* Sample containers shall be taken at random from each lot as follows:

Lot size	*Sample size*
1-99	1
100-500	2
More than 500	3

A specimen shall be removed from each container in the sample and placed in a clean, dry container labeled to identify the lot and the container from which it was taken. Each specimen shall be tested as specified in 4.6.

4.5 **Examination.** Sample containers of thionyl chloride shall be examined for compliance with level A preparation for delivery requirements in accordance with the classification of defects and with MIL-STD-105.

4.5.1 *Classification of defects.*

4.5.1.1 *Preparation for delivery (section 5).*

Categories	*Defects*
Critical:	None defined
Major:	*AQL 1.5 percent defective*
101	Unit, intermediate, or shipping container not as specified
102	Cushioning of bottles not as specified
103	Fiberboard fillers not as specified
104	Unit, intermediate, or shipping container damaged
105	Closure of unit, intermediate, or shipping container incorrect
106	Marking of unit, intermediate, or shipping container incorrect, illegible, or missing

4.6 **Tests.** Distilled water and reagent grade chemicals shall be used throughout tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

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4.6.1 *Iron.*

4.6.1.1 *Preparation of standard iron solution.* Dissolve about 2.0 grams (g.) of ferric chloride ($FeCl_3.6H_2O$) in a liter of water containing 10 milliliters (ml.) of concentrated hydrochloric acid. Determine the exact ferric iron content of this solution by a standard method. From a burette run enough of this solution into a 1-liter volumetric flask to make a concentration of 30.0 milligrams (mg.) of iron in 1 liter (about 75 ml. will be required). Add 10 ml. of concentrated hydrochloric acid and dilute to the mark. One ml. of this solution contains 0.03 mg. of iron.

4.6.1.2 *Procedure.* By means of a weighing pipette, transfer about 3.50 g. of specimen to a 150-ml. beaker. Add 2 ml. concentrated sulfuric acid, place on a hotplate in a hood, and carefully evaporate the thionyl chloride. To the residue, add 1 ml. concentrated nitric acid and heat to fumes of sulfur trioxide. Cool, cautiously add 20 ml. water and 2 ml. of 6*N* hydrochloric acid, and boil for a few minutes. Cool, transfer to a 250-ml. volumetric flask, and dilute to the mark. Pipette a 10.0-ml. aliquot into a 50-ml. Nessler tube, add 10.0 ml. of 5-percent potassium thiocyanate solution, dilute to the mark, and compare with colorimetric standards freshly prepared as follows: Mix measured volumes of standard iron solution (see 4.6.1.1), 1.0 ml. of 1 to 10 sulfuric acid, and 10 ml. of 5-percent potassium thiocyanate solution in a 50-ml. Nessler tube, and dilute to the mark. Record the estimated weight of iron in the sample.

$$\text{Percent iron as } FeCl_3 = \frac{\text{estimated mg. of iron} \times 227}{\text{mg. of specimen in aliquot}}$$

4.6.2 *Chlorides of sulfur.*

4.6.2.1 *Preparation of color standards.* Prepare 1500 ml. of thionyl chloride which is substantially free from chlorides of sulfur (see 6.7). Prepare about 200 ml. of a stock solution containing 10 percent by weight of sulfur monochloride in the "purified" thionyl chloride. Using this solution, prepare 100 ml. each of a series of concentrations of sulfur monochloride in the "purified" thionyl chloride, from 0.0 percent to 2.0 percent in steps of 0.2 percent. Place each solution so prepared in a 4-oz. glass-stoppered bottle made of clear, flint glass and label each bottle with the percent of sulfur monochloride it contains. Add 1 to 2 mg. of ferric sulfate to the contents of each bottle (see 6.11). Into the solution in each bottle, pass a rapid stream of dry chlorine gas for 5 minutes at room temperature. Seal and date the bottles. Check the standards at regular intervals and replace when necessary (see 6.8).

4.6.2.2 *Procedure.* Distill 150 ml. of the specimen in a stream of dry chlorine gas to near dryness. The distillate contains all the chlorides of sulfur as sulfur dichloride and is iron-free. Place 100 ml. of the well-mixed distillate in a 4-oz bottle similar to those used in the preparation of the standards (see 4.6.2.1). Add 1 to 2 mg. of ferric sulfate to the contents of the bottle (see 6.11). Pass a rapid stream of dry chlorine into the bottle for 5 minutes at room temperature. Stopper, let stand at least 3 hours, and compare the color developed with those of the standards. Record the estimated percentage of sulfur monochloride in the sample (see 6.9).

4.6.3 *Sulfuryl chloride.* Place 50 ml. of sample in a 100-ml. distillation flask of the Engler type. Close the neck with a cork and distill on a boiling water bath. Condense the vapors with a water-cooled glass condenser. Continue the distillation to dryness. Agitate the distillate to insure uniform composition. Weigh 1.500 to 2.000 g. of the distillate in a thin glass bulb. Place the bulb in a 300-ml. glass-stoppered Erlenmeyer flask containing 150 ml. of water. Stopper and break the bulb by shaking. Continue shaking for 1 minute. Allow the flask to stand for 1 hour with occasional shaking.

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Remove and wash down the stopper; cover the flask with a watchglass. Bring to a boil and maintain ebullition until the sulfur dioxide has been completely driven off. Filter through paper into a beaker, wash, and add 1 ml. of saturated aqueous solution of picric acid. Bring to a boil, add at one stroke, while stirring, 4.0 ml. of 10 percent barium chloride solution, boil gently for 10 minutes, filter through an ignited, tared Gooch crucible. Dry in an oven at 105° to 110°C., ignite at a dull red heat, cool in a desiccator, and weigh.

$$\text{Percent } SO_2Cl_2 = \frac{\text{weight } BaSO_4 \times 57.8}{\text{weight of sample}}$$

4.6.4 *Sulfur dioxide* Place 100 ml. of sample in a 4-oz. glass-stoppered bottle, add 1 g. of flowers of sulfur, and let stand for 1 hour, with occasional shaking. (This reduces any SCl_2 to the higher boiling S_2Cl_2 and prevents interference of SCl_2 in the SO_2 determination). Set up the apparatus as shown in figure 1 (see 6.10). Pass a rapid stream of nitrogen from cylinder A through the sulfuric acid washing bottle B into the apparatus for 5 minutes to clear the system; then reduce the rate of flow to 1 bubble per second. Charge absorber tube E with 75.0 ml. of approximately 1.0N sodium hydroxide and add sufficient water to bring the level to about 1 inch from the top of the test tube. Disconnect tube D and open the apparatus at inlet J. Measure 50 ml. (equal to 83 g.) of the specimen in a 50-ml. graduate (graduated to deliver 50 ml. and transfer quickly to flack C. Immediately connect the apparatus to tube D, then to the gas inlet at J, and readjust the nitrogen stream to 1 bubble per second. Heat flask C gently until the thionyl chloride refluxes slowly, increase the rate of nitrogen flow to 2 bubbles per second, and continue refluxing for 3 to 5 minutes. Discontinue heating and increase the rate of flow of nitrogen to prevent sucking-back of the absorbing solution as flask C cools. Transfer the absorbing solution quantitatively to a 250-ml. volumetric flask and dilute to the mark with water and mix thoroughly. Pipette a 50-ml. aliquot into a 250-ml. glass-stoppered flask, add 10 ml. of 3 percent hydrogen peroxide solution, and titrate with standard 0.5*N* sulfuric acid, using methyl red as indicator. Retain the titrated solution for determination of chloride. Run a blank, using a 50-mil. aliquot of a 250-ml. dilution of 75 ml. of the 1.0 *N* sodium hydroxide and 10 ml. of a 3-percent hydrogen peroxide solution as in the preceding titration. Call the difference between the two titrations "ml. H_2SO_4." To each of the above solutions add 1 ml. of 6*N* nitric acid and 3 to 5 g. of precipitated calcium carbonate. Add 1 ml. of 4.25M potassium chromate indicator solution and titrate with 0.1*N* silver nitrate by the Mohr method. Call the difference between blank and sample titrations "ml. $AgNO_3$."

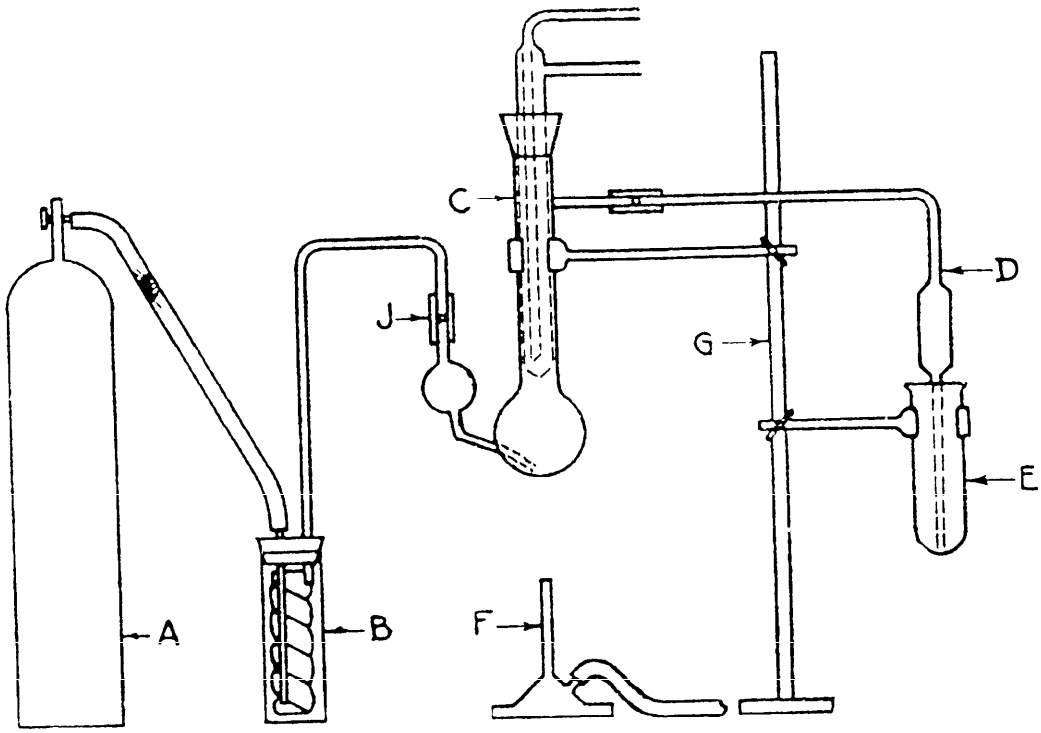
$$\text{Percent } SO_2 = \frac{(\text{"ml } H_2SO_4\text{"} \times A \times 3.2) - (\text{"ml } AgNO_3\text{"} \times B \times 6.4)}{\text{weight of specimen} \times 0.2}$$

where: A = normality of H_2SO_4 solution, and
B = normality of $AgNO_3$ solution.

4.6.5 *Purity.*

4.6.5.1 *Purity by total sulfur.* Weigh about 1.750 g. of specimen in a thin glass bulb. Place the bulb in a heavy-walled, 500-ml., glass-stoppered Erlenmeyer flask containing 13 g. of sodium hydroxide dissolved in 150 ml. of water. Stopper the flask and break the bulb by shaking. Continue shaking until the mist disappears and let stand 15 minutes with occasional shaking. Remove stopper and wash down stopper and flask. Crush stem and fragments of the bulb with a heavy glass rod. Remove and rinse the rod. Add 10 ml. of 30-percent hydrogen peroxide (or an equivalent quantity of a weaker strength hydrogen peroxide). A piece of platinum, e.g. foil or crucible cover, may be added as a catalyst if desired. Digest about 40 minutes on the steam bath, neutralize with 6*N* hydrochloric acid, and add 2 to 5 ml. in excess. Add sufficient bromine water to color the solution light yellow and filter into a 500-ml. volumetric flask. Wash with water, cool to room

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LEGEND
A - Nitrogen Cylinder (with reducing control valve)
B - Sulfuric Acid Washing Bottle
C - Koch Flask, 300 ml. capacity
D - Delivery Tube
E - Test Tube containing Absorbent Solution
F - Micro-burner
G - Support Stand and Clamps
J - Disconnect Sleeve

APPARATUS
SO_2 DETERMINATION

FIGURE 1.

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temperature, dilute to the mark, and mix thoroughly. Pipette a 100-ml. aliquot into a 400-ml. beaker; add one or two drops of ammonium hydroxide and 2 ml. of a saturated solution of picric acid. Dilute to 300 ml. and heat to boiling. Add rapidly 10 ml. of a 10-percent barium chloride solution, boil gently for 3 to 5 minutes, filter through a tared, ignited, porous-bottom porcelain crucible or a Gooch crucible, and wash free from chlorides. Dry in an oven at 105° to 110°C., ignite in a muffle furnace at a dull red heat, cool in a desiccator, and weigh.

$$\text{Percent } SOCl_2 \text{ based on total sulfur} = \left[\frac{\text{wt. } BaSO_4 \times 13.74}{\text{wt. of specimen} \times 0.2} - (A + B)\right] \times 3.71$$

where: A = percent chlorides of sulfur × 0.47, and
B = percent SO_2Cl_2 × 0.238

4.6.5.2 *Purity by difference.* Calculate the purity by difference as follows:

Percent purity by difference = 100 — (percent SO_2Cl_2 + percent S_2Cl_2)

4.7 **Acceptance/rejection criteria.** If any specimen fails to meet the test requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* Thionyl chloride, in the unit quantity specified (see 6.2), shall be packaged in accordance with the requirements for type I, class 1, inclosure b of PPP-C-300.

5.1.2 *Level C.* Thionyl chloride shall be packaged in accordance with the manufacturer's commercial practice.

5.2 **Packing.** Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Levels A, B, and C.* Thionyl chloride shall be packed in accordance with the application requirements of PPP-C-300.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with ICC regulations and MIL-STD-129.

6. NOTES

6.1 **Intended use.** The material covered by this specification is intended for use as a chemical reagent.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Unit quantity required.

6.3 **Batch.** A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 **Military use.** Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard. Personnel of the military departments are requested to refer to these documents for guidance.

6.5 **Supersession data.** This specification includes the requirements of Military Specification MIL-T-1045A, Thionyl Chloride, Technical, dated 10 May 1951.

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6.6 **Significant places.** For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation: E29).

6.7 **Purification.** The best grade of thionyl chloride furnished by Eastman Kodak Company, Rochester, New York, Catalog No. 246, has been found sufficiently pure for use without further purification. It has a boiling range of 75° to 76°C. and is nearly water white. Most commercial grades of thionyl chloride may be purified sufficiently to permit use in sulfur monochloride color standards by distillation from linseed oil or from phosphorus tribromide containing flowers of sulfur; the first and last fractions should be rejected. However, regardless of its source, "purified" thionyl chloride must be tested before use by passing a rapid stream of chlorine through the material for 5 minutes. The thionyl chloride will turn a clear yellow due to dissolved chlorine. However, no brownish tint or other coloration developed after 3 hours, sufficient to interfere with the tests, should be permitted. The "purified" thionyl chloride should be colorless when 100 ml. are placed in a 4-oz. glass bottle prior to chlorination.

6.8 **Standards.** Standards prepared as indicated in 4.6.2.1 are sufficiently stable to last 6 months before replacement. However, the standards should be checked at regular intervals.

6.9 **Chlorination.** If the original sample contains less than 0.02 percent ferrous chloride and is free from any brown or greenish discoloration, it may be chlorinated directly without previous distillation in a stream of chlorine.

6.10 **Figure 1.** The flask and condenser (C) constitute a Koch carbon apparatus, except that the flask has a capacity of 300 ml. instead of 500 ml. as shown in the catalogs of the Scientific Glass Company, Bloomfield, New Jersey (Catalog No. J865), and the Ace Glass Company, Vineland, New Jersey (Catalog No. 6250).

6.11 **Catalyst.** A definite trace of iron is added to the standards and to the sample to serve as a catalyst to speed up the conversion of sulfur monochloride to sulfur dichloride.

6.12 **Transportation description.** Transportation description and minimum weights applicable to this commodity are:

Rail:

Chemicals, not otherwise indexed by name.

Carload minimum weight 24,000 pounds, Subject to Rule 34, Uniform Freight Classification.

Motor:

Chemicals, not otherwise indexed.

Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY INTEREST:
Army—C O
Navy—Sh
Air Force—MAA

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